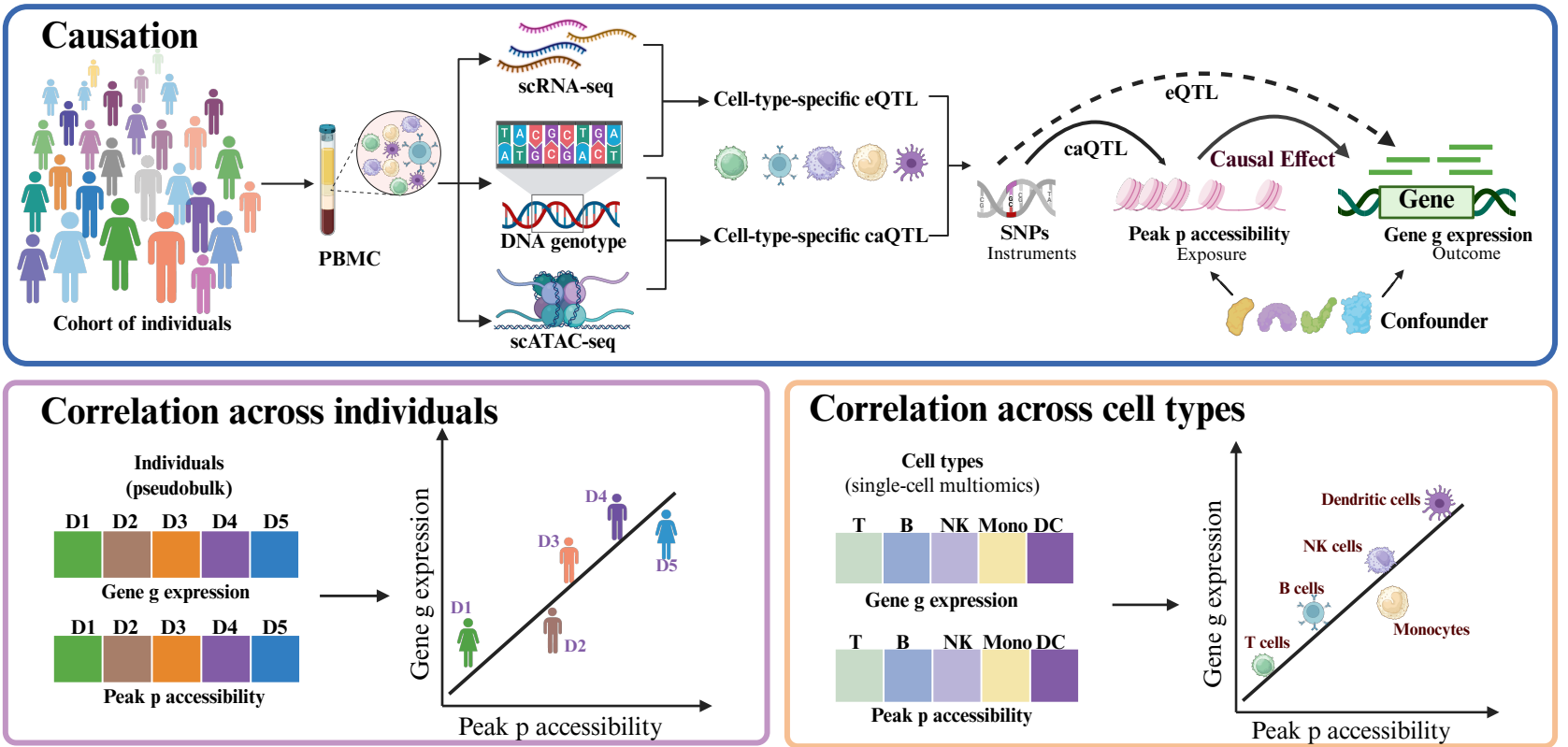

Causation
Cohort of individuals
PBMC
scRNA-seq
DNA genotype
scATAC-seq
Cell-type-specific eQTL
Cell-type-specific caQTL
eQTL
caQTL
Causal Effect
SNPs
Instruments
Peak p accessibility
Exposure
Gene
Gene g expression
Outcome
Confounder
Correlation across individuals
Individuals
(pseudobulk)
D1 D2 D3 D4 D5
Gene g expression
D1 D2 D3 D4 D5
Peak p accessibility
Gene g expression
Peak p accessibility
Correlation across cell types
Cell types
(single-cell multiomics)
T B NK Mono DC
Gene g expression
T B NK Mono DC
Peak p accessibility
Gene g expression
Peak p accessibility
Dendritic cells
NK cells
B cells
Monocytes
T cells